



Module Presenter's Manual

for

Programming in C#

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Ver. 1.0

Amendment Record

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1.0	April 2024	New	-

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1. Introduction

At the end of this module, students will be able to:

- Understand how C# fits into the .NET Platform
- Analyze the basic structure of a C# application and be able to document, debug, compile, and run simple application
- Create, name, and assign values to variables
- Use common statements to implement flow control, looping, and exception handling
- Create methods (functions and subroutines) that can return values and take parameters
- Explain the basic concepts and terminology of object-oriented programming
- Create, initialize, and destroy objects in a C# application
- Build new C# classes from existing classes
- Create self-contained classes and frameworks in a C# application
- Define operators, use delegates, and add event specifications
- Explore the new features of C# 10.0
- Explain .NET Core Development
- Build Cross-Platform Mobile Apps Using .NET MAUI
- Explore GUI and Desktop Apps with C#

2. Information on Session Allocation

Module	Online Hours	Self Study Hours
Programming in C#	40	12

Throughout this Presenter's Manual, the module **Programming in C#** will be referred to as **PC#**.

3. Module Deliverables on OnlineVarsity

To aid the teaching process, following are the deliverables:

Resources available on Onlinevarsity:

Feature - Description/Functionality
Ebook -Student reading material in the form of PDF.
Download eBook - Student has the option to download the subject related e-book and read offline.
Practice 4 Me - Student can test and evaluate their understanding of module related topics.
Work Assignments - Student can solve scenario based lab assignments (Hands-on). The faculty will evaluate and give their feedbacks.
References - Student can access additional subject related material for reading.

4. Week-wise Session Schedule

- A session is of 2 hours duration.

Week	Day 1	Day 2		Day 3	Day 4	
1	Session 1 TL1	Session 2 TL2		Session 3 TL3	Session 4 TL4	
2	Session 5 TL5	Session 6 TL6		Session 7 TL7	Session 8 TL8	
3	Session 9 TL9	Session 10 TL10	S1-S3	Session 11 TL11	Session 12 TL12	
4	Session 13 TL13	Session 14 TL14		Session 15 TL15	Session 16 TL16	
5	Session 17 TL17	Session 18 TL18		Session 19 TL19	Session 20 TL20	S4-S6

TL: Online Session

S: Self Study Session (over and above the module duration)

5. Session Coverage

Session No.	Session Title	Session Details	Deliverables' Mapping
1	PC# – TL1	All the topics as listed below from Session 1 and Session 2 of <i>Developing Applications with C#</i> book should be covered in this session. <u>Session 1 – Getting Started with C#</u> ➤ Define and describe the .NET Framework ➤ Explain Components of .NET 7.0 ➤ Explain C# language features ➤ Identify various editions of Visual Studio 2022 ➤ Outline the differences between .NET Framework 4.8.1 and .NET 7.0 ➤ Explain the process of running a program on .NET 7.0 <u>Session 2 – Basic Building Blocks in C#</u> ➤ Define and describe variables and data types in C# ➤ Explain comments and XML documentation ➤ Define and describe constants and literals ➤ Describe constant interpolated strings ➤ List the keywords and escape sequences ➤ Explain input and output ➤ Explain different types of statements ➤ Explain the use of different types of operators	<u>Developing Applications with C#</u> SG - Session 1 & 2 XP - Session 1 & 2 TG - Session 1 & 2
2	PC# – TL2	All the topics as listed below from Session 3 of <i>Developing Applications with C#</i> book should be covered in this session. <u>Session 3 – Programming Constructs and Arrays</u> ➤ Explain selection constructs ➤ Describe loop constructs ➤ Explain jump statements in C# ➤ Define and describe arrays ➤ Explain different types of arrays ➤ Describe the Array class	<u>Developing Applications with C#</u> SG - Session 3 XP - Session 3 TG - Session 3
3	PC# – TL3	The Try It Yourself questions of Session 1 to Session 3 of <i>Developing Applications with C#</i>	<u>Developing Applications with C#</u>

Session No.	Session Title	Session Details	Deliverables' Mapping
		book should be covered in this session.	Session 1-3
4	PC# – TL4	All the topics as listed below from Session 4 of <i>Developing Applications with C#</i> book should be covered in this session. <u>Session 4 – Classes and Methods in C#</u> ➤ Explain classes and objects ➤ Define and describe methods ➤ List the access modifiers ➤ Explain method overloading ➤ Define and describe constructors and destructors	<u>Developing Applications with C#</u> SG - Session 4 XP - Session 4 TG - Session 4
5	PC# – TL5	All the topics as listed below from Session 5 of <i>Developing Applications with C#</i> book should be covered in this session. <u>Session 5 – Inheritance and Polymorphism</u> ➤ Define and describe inheritance ➤ Explain method overriding ➤ Define and describe sealed classes ➤ Explain polymorphism	<u>Developing Applications with C#</u> SG - Session 5 XP - Session 5 TG - Session 5
6	PC# – TL6	All the topics as listed below from Session 6 of <i>Developing Applications with C#</i> book should be covered in this session. <u>Session 6 – Abstract Classes and Interfaces</u> ➤ Define and describe abstract classes ➤ Explain interfaces ➤ Compare abstract classes and interfaces	<u>Developing Applications with C#</u> SG - Session 6 XP - Session 6 TG - Session 6
7	PC# – TL7	The Try It Yourself questions of Session 4 to Session 6 of <i>Developing Applications with C#</i> book should be covered in this session.	<u>Developing Applications with C#</u> Session 4-6
8	PC# – TL8	All the topics as listed below from Session 7 of <i>Developing Applications with C#</i> book should be covered in this session. <u>Session 7 – Properties, Indexers, and Record Types</u> ➤ Define properties in C# ➤ Explain properties, fields, and methods	<u>Developing Applications with C#</u> SG - Session 7 XP - Session 7 TG - Session 7

Session No.	Session Title	Session Details	Deliverables' Mapping
		➤ Explain indexers ➤ Describe init only setters and record types	
9	PC# – TL9	All the topics as listed below from Session 8 of <i>Developing Applications with C#</i> book should be covered in this session. <u>Session 8 – Namespaces and Exception Handling</u> ➤ Define and describe namespaces ➤ Define and describe exceptions ➤ Explain the process of throwing and catching exceptions ➤ Explain nested try and multiple catch blocks	<u>Developing Applications with C#</u> SG - Session 8 XP - Session 8 TG - Session 8
10	PC# – TL10	All the topics as listed below from Session 9 of <i>Developing Applications with C#</i> book should be covered in this session. <u>Session 9 – Events, Delegates, and Collections</u> ➤ Explain delegates ➤ Explain events ➤ Define and describe collections	<u>Developing Applications with C#</u> SG - Session 9 XP - Session 9 TG - Session 9
	S1-S3	All the Assignments of Sessions 1 to 9 of <i>Developing Applications with C#</i> course should be covered in these sessions	<u>Developing Applications with C# OnlineVarsity</u>
11	PC# – TL11	The Try It Yourself questions of Session 7 to Session 9 of <i>Developing Applications with C#</i> book should be covered in this session.	<u>Developing Applications with C#</u> Session 7-9
12	PC# – TL12	All the topics as listed below from Session 10 of <i>Developing Applications with C#</i> book should be covered in this session. <u>Session 10 – Generics and Iterators</u> ➤ Define and describe generics ➤ Explain creating and using generics ➤ Explain iterators	<u>Developing Applications with C#</u> SG - Session 10 XP - Session 10 TG - Session 10
13	PC# – TL13	All the topics as listed below from Session 11 of <i>Developing Applications with C#</i> book should be covered in this session. <u>Session 11 – GUI and Connectivity with SQL Database</u>	<u>Developing Applications with C#</u> SG - Session 11 XP - Session 11 TG - Session 11

Session No.	Session Title	Session Details	Deliverables' Mapping
		➤ Define GUI ➤ Identify tools for creating GUI applications ➤ Explain Forms and Controls ➤ Elaborate on the implementation of GUI in C# ➤ Explain database connectivity ➤ Describe SQL Server Database Connectivity using C# ➤ List SQL commands ➤ Explain insert, update, and delete operations in an SQL database	
14	PC# – TL14	All the topics as listed below from Session 12 of <i>Developing Applications with C#</i> book should be covered in this session. <u>Session 12 – Advanced Concepts in C#</u> ➤ Describe anonymous methods and extension methods ➤ Explain anonymous types, partial types, and nullable types ➤ Describe system-defined generic delegates ➤ Define lambda expressions ➤ Explain query expressions ➤ Explain parallel and dynamic programming	<u>Developing Applications with C#</u> SG - Session 12 XP - Session 12 TG - Session 12
15	PC# – TL15	The Try It Yourself questions of Session 10 to Session 12 of <i>Developing Applications with C#</i> book should be covered in this session.	<u>Developing Applications with C#</u> Session 10-12
16 & 17	PC# – TL16 & TL17	All the topics as listed below from Session 13 of <i>Developing Applications with C#</i> book should be covered in this session. <u>Session 13 – Building Cross-Platform Mobile Apps Using .NET MAUI</u> ➤ Describe the .NET MAUI architecture ➤ List the features of MAUI ➤ Explain how to install and run the application	<u>Developing Applications with C#</u> SG - Session 13 XP - Session 13 TG - Session 13
18 & 19	PC# – TL18 &	All the topics as listed below from Session 14 of <i>Developing Applications with C#</i> book should	<u>Developing Applications with C#</u>

Session No.	Session Title	Session Details	Deliverables' Mapping
	TL19	be covered in this session. <u>Session 14 – .NET Development and the Future</u> ➤ Explain what .NET Core is ➤ Explain the history of .NET Core and .NET ➤ Identify differences between .NET Framework and .NET Core ➤ Explain .NET 7.0 and its key features ➤ Describe how to create an application with .NET 7.0 and VS 2022 ➤ Outline .NET 8.0 and the Future	SG - Session 14 XP - Session 14 TG - Session 14
20	PC# – TL20	The Try It Yourself questions of Session 13 and Session 14 of <i>Developing Applications with C#</i> book should be covered in this session.	<u>Developing Applications with C#</u> Session 13 & 14
	S4-S6	All the Assignments of Sessions 10 to 14 of <i>Developing Applications with C</i> course should be covered in these sessions.	<u>Developing Applications with C# OnlineVarsity</u>

6. Library References

➤	Programming C# 10 by Ian Griffiths
➤	C# 10 and .NET 6 – Modern Cross-Platform Development: Build apps, websites, and services with ASP.NET Core 6, Blazor, and EF Core 6 using Visual Studio 2022 and Visual Studio Code 6th ed. Edition by Mark J. Price
➤	Pro C# 10 with .NET 6: Foundational Principles and Practices in Programming 11st ed. Edition by Andrew Troelsen, Phil Japikse

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